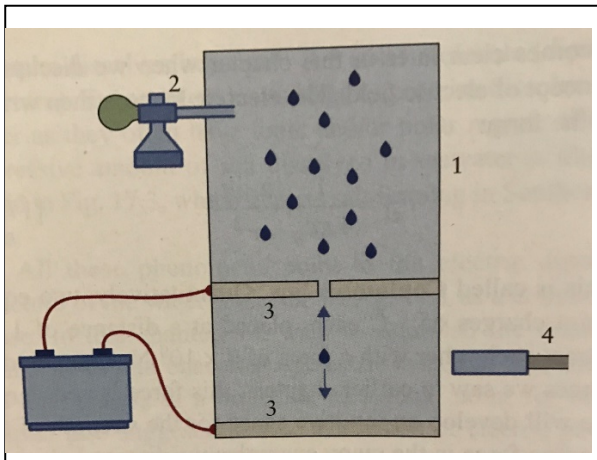
Question 1 **11 marks**

(a) Two point charges attract each other with an electric force of magnitude F. If we triple one of the charges and increase the distance between the charges by a factor of 3, what does the magnitude of the force between the particles become?

The charge is tripled, so F increases by a factor of 3. But the distance is reduced to 1/3, so the force is reduced by a factor of 9. Hence Ans = F/3

(b) In Robert Millikan's experiment, oil mist, consisting of microscopic oil droplets, is sprayed into a chamber (1). Many of the oil droplets carry charges due to the friction of the oil when leaving the metallic nozzle of the vaporizer (2). The droplets sink due to their weight. It is possible to levitate individual oil droplets between two charged metallic plates (3) that form a horizontal, parallel plate arrangement. A mechanical equilibrium can be observed with a microscope (4) and is due to the balance between the weight and the electric force acting upward, as indicated on one lower oil droplet by two arrows. From this experiment the electric charge on the oil droplet is quantified and is always found to be a multiple of the elementary charge. The plates (3) are separated by a distance of 1mm. It is found that when the voltage on the plates is increased to 150V, oil droplets of radius 1.4μm and density 0.85 g/cm^3 are levitated in the chamber. The volume of a sphere is given by $V = \frac{4}{3}pr^3$



(i) What is the electric field between the plates?
(ii) What is the gravitational force on the droplet?
(iii) What is the electric force on the droplet?
(iv) Assume that the top plate (3) is positively charged. What is the sign of the charge on the droplet?
(v) What is the magnitude of the charge on the droplet?
(vi) How many elementary charges does the droplet carry?

(i) Electric Field between the plates = 150/1e-3 = 1.5e5 V/m
(ii) Volume of droplet is (4/3) π r^3 . r = 1.4e-6m, hence volume = 11.44e-18m^3,
Mass = 0.85e3 kg/m^3*11.44e-18 = 9.76e-15kg; F = mg = 9.568e-14N
(iii) 9.568e-14N
(iv) Negative
(v) F=qE, hence q = 9.568e-14/1.5e5 = 6.38e-19C,
(vi) number of charges = 4.

[2 + (2 + 2 + 1 + 1 + 1 + 2) = 11 marks]

Question 2 **6 marks**

(a) A parallel plate capacitor has an initial capacitance of C = 10 pF. It is charged up using a 9V battery.

i. What is the electrical potential energy stored in the capacitor?

A piece of glass is now inserted between the plates. The dielectric constant of glass is 4.7

ii. What is the new capacitance with the glass inserted?

iii. What is the energy stored in the capacitor after the glass is inserted

iv. Compare this energy with the energy originally stored in the capacitor before the glass was inserted. If it is less, where has the energy gone? If it is more, where has the extra energy come from?

[1 + 1 + 2 + 2 = 6 marks]

(i) ½ CV^2 = ½ * 10pF*81 = 405 pJ
(ii) C = 10*4.7 = 47 pF
(iii) The voltage decreases by this factor, so V across the plates is now 1.91
And so energy is .5*47*1.91*1.91 = 85.7pJ
(iv) Energy is less. It has been lost in heat as the the glass was inserted into the capacitor.

Question 3 **6 marks**

(a) A mass spectrometer is used to separate isotopes. The beam emerges with a speed of 250 km/s and the magnetic field in the mass selector is 2T.

(i) What is the radius of the path of an ion of mass m and unit charge in the mass selector?
(ii) What is the distance between the collectors for 235U and 238U?
(iii)What is the advantage of using a mass selector with a very high magnetic field?

[2 + 2 + 2 = 6 marks]

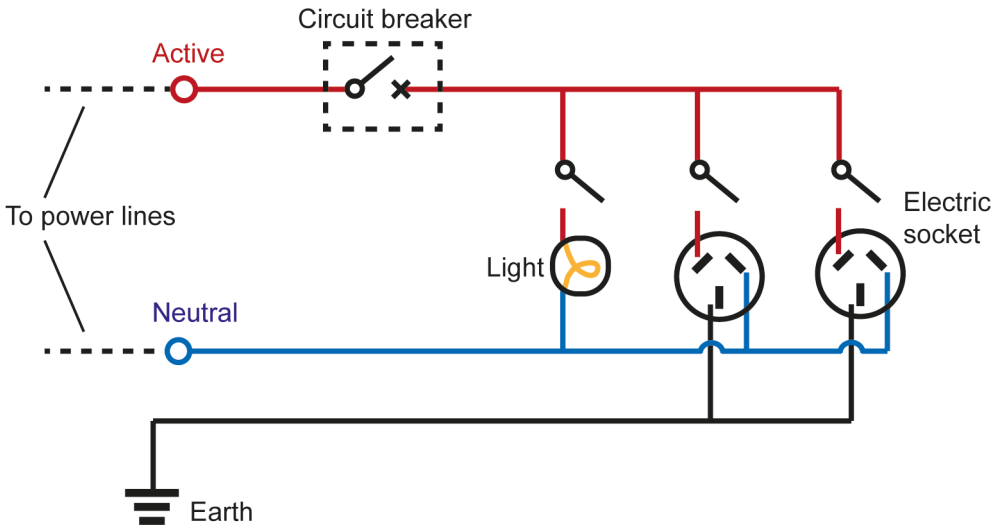
(i) $MV^2/R = Bqv$ so r = mv/BQ

(ii) r (per amu) = 1.67e-27*250e3/(2*1.6e-19) = 0.0013m = 1.3mm.
Since there are 3 mass units between 235U and 238U the distance will be 7.8mm
(ii) The separation will be smaller and so the whole machine can be more compact.

Question 4
(has been removed, as it was not covered in the lecture)

Question 5 **11 marks**

(a) A simplified model of a household electric circuit is shown below.



(i) Explain why lights and electrical appliances are connected in parallel.

(ii) What happens to the current in the circuit as more lights are switched on and more appliances are plugged in and switched on?

(iii) What is the role of the earth pin on appliances such as refrigerators and washing machines?

In the questions that follow, assume a constant voltage power supply of 230 V (i.e. ignore the fact that mains power is AC).

(b) An electric toaster dissipates 1200 W of power.

(i) Find the current in the toaster element.

(ii) What is the resistance, R, of the toaster element?

(iii) The toaster element consists of a wire made from Nichrome that has a resistivity of 1.3×10^{-6} Ω.m. If the wire has a circular cross-section with a diameter of 270 □m, what is the length of this wire?

[6 + 5 = 11 marks]

(a) (i) Appliances are connected in parallel to ensure that the potential difference across each device is independent of how many other devices are switched on.

(ii) The current increases.

(ii) The earth pin electrically connects the conducting exterior of the applicance to earth at the residence. This ensures that the exterior of the device is always at ground potential. This means that if there is a fault with the applicance, users are protected from electric shock.

(b) (i)

$$P = IV$$

$$I = \frac{1200}{230} = 5.22A$$

(ii)

$$V = IR$$

$$R = \frac{V}{I} = \frac{230}{5.22} = 44.1\Omega$$

(iii)

$$R = \frac{\rho l}{A}$$

$$l = \frac{RA}{\rho}$$

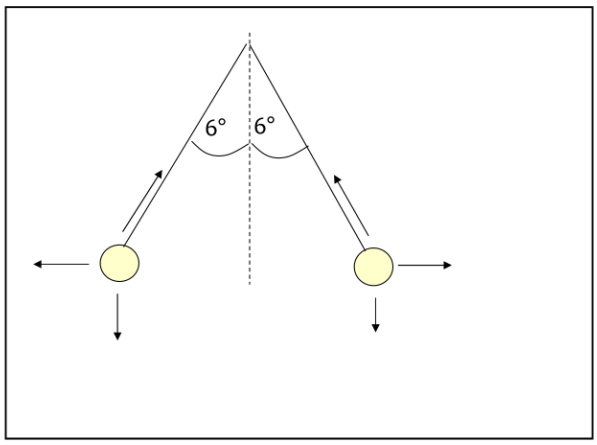
Here $A = \pi r^2 = \pi \times (270 \times 10^{-6}/2)^2 = 5.725 \times 10^{-8} m^2$

So $l = \frac{44.1 \times 5.725 \times 10^{-8}}{1.3 \times 10^{-6}} = 1.9m$

Question 6 **13 marks**

(a) Can electric field lines ever cross? Explain your answer.

(b) Two pith balls, mass 0.25 g each, hang by 25 cm threads (assume threads have no mass). The angle of separation is 12° (see figure)



(i) What is the charge on the balls? Assume that the charge on each ball is the same.

(ii) If the balls are positively charged, how many electrons have been removed from each ball?

(c)

(i) What is an electric dipole? Include a diagram in your description.

(ii) In order to melt pure NaCl into liquid requires very high temperatures (approximately 800^{0}C). And yet, NaCl will readily dissolve in water at room temperature. Explain how water can so readily break the Na-Cl bond. Your answer should include a consideration of ion-dipole interactions.

[2 + (5 + 1) + (2 + 3) = 13 marks]

6(a). No. Electric field lines represent the direction of the force at the respective point in space. If two lines would cross, an ambiguity about the direction of the electric force at the intersection would occur. 2 marks

6(b) (i) We focus on one of the charges and analyze the forces acting on it. If you do the free-body diagram for the charge on the right, three forces act on it: the gravitational force directed down of magnitude $F_g = mg$, the tension T directed along the string at an angle of θ with respect to the vertical, and the electrostatic repulsion F_{el} directed to the right, and of magnitude:

$$F_{el} = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{d^2}, \qquad \textbf{(1)}$$

where q is the charge of each sphere and d is the separation between the spheres. Since the sphere is hanging at equilibrium, the horizontal components and vertical components will balance. This means that the horizontal component of the tension, directed to the left, will balance the electrostatic repulsion, directed to the right:

$$T\sin\theta = F_{el} \qquad \textbf{(2)}$$

Also, the vertical component of the tension, directed up, will balance the gravitational force, directed down:

$$T\cos\theta = mg \qquad \textbf{(3)}$$

We can take now the ratio of Eq. [2] to Eq. [3] to cancel the unknown tension and obtain:

$$\tan\theta = \frac{F_{el}}{mg} = \frac{q^2}{4\pi\varepsilon_0 d^2 mg}. \qquad \textbf{(4)}$$

Given the geometry of Figure 16.24, the length L of the string is related to the separation between the spheres d according to:

$$d = 2L\sin\theta. \qquad \textbf{(5)}$$

We can substitute the value of d from Eq. [5] into Eq. [4] and isolate q, yielding:

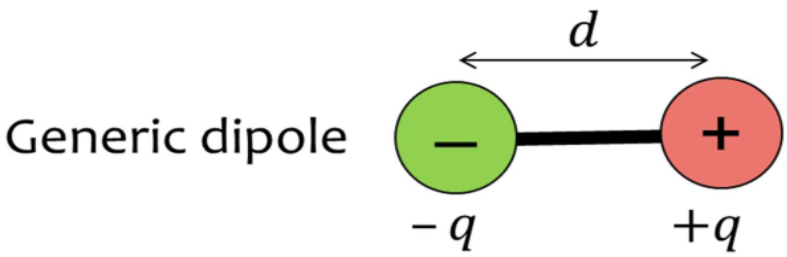
$$q = \sqrt{\frac{16\pi\varepsilon_0 L^2 mg\sin^3\theta}{\cos\theta}} = 8.84\times10^{-9}\,\text{C}, \qquad \textbf{(6)}$$

where L = 25 cm, m = 0.25 g, θ = 6.0°, and the permittivity is $\varepsilon_0 = 8.85 \times 10^{-12}$ C²/(N m²). In more appropriate units, Eq. [6] tells us that the charge in each sphere must be q = 8.8 nC.

6b (ii) . Ans = 8.8e-9/1.6e-19 = 55E9 electrons **1 mark**

6c.

(i) A configuration of two point charges of equal magnitude and opposite sign separated by a fixed distance is called an electric dipole.



2 marks

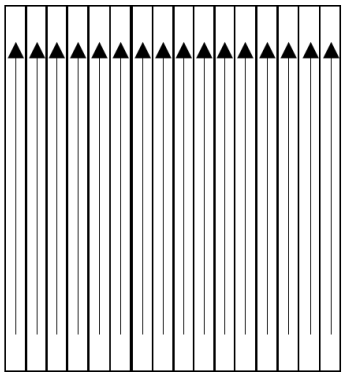
(ii) See case study 16.1, p448, Zinke, third edition.

3 marks

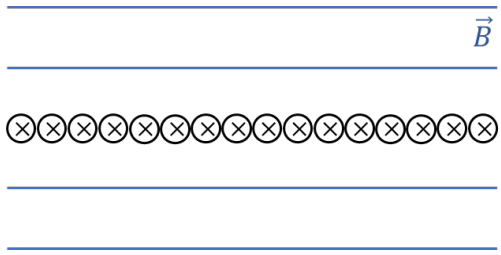
Question 7 **6 marks**

It can be shown that a very large sheet of flat, densely packed wires each carrying a current, I, produces magnetic field lines $\vec{B}$ that are parallel to the surface of the sheet. Lines of magnetic field $\vec{B}$ are shown as blue lines in the right (side view) figure, although the direction is not indicated. The arrows in the top view denote the direction of the current and the symbol ⊗ in the side view denotes current directed into the page.

Top view



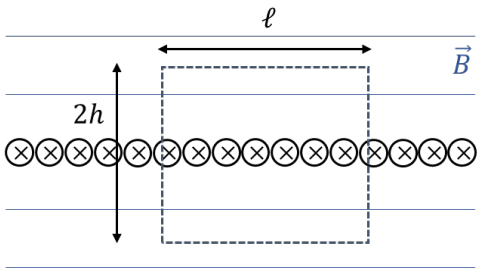
Side view



(a) With reference to the side view in the figure above will $\vec{B}$ in the region of space *above* the wires be directed right or left?

(b) With reference to the side view in the figure above will $\vec{B}$ in the region of space *below* the wires be directed right or left?

Assume that the full width of the array of wires is W and there are N wires in total, each carrying a current I. Consider the Amperian loop shown as a dashed line in the figure on the right. The vertical sides have length 2h and the top and bottom sides have length l. The top side is the same distance, h, from the wires as the bottom side. Assume that B does not vary in magnitude along the top and bottom sides since the array of wires is large.



(c) Write down an expression for the total current passing through the loop in terms of I, N, W and l.

(d) Briefly explain why would you expect the magnetic field on the top side to be the same as that along the bottom side?

[1 + 1 + 2 + 2 = 6 marks]

(a) Right

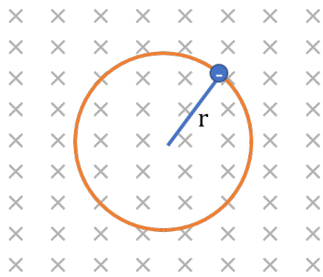
(b) Left

c) $I_{enclosed} = I * \frac{l}{W} * N = \frac{IlN}{W}$

(d) Symmetry. Since the sheet of current is very wide, there is no difference between lines located equidistant from the sheet.

Question 8 **3 marks**

(a) A particle with charge, q, moves with speed v in a circular orbit of radius r in a uniform magnetic field of magnitude, B, directed perpendicular to the plane of the orbit. In the figure below, this corresponds to a field directed into the page.



i. According to the figure above, if the particle is negatively charged, does it move in a clockwise or anti-clockwise direction?

ii. The centripetal (central) force on a particle of mass m, moving in a circle of radius r at speed v is given by:

$$F_{central} = \frac{mv^2}{r}.$$

Assuming that the magnetic force provides this central force, show that the orbit radius is given by:

$$r = \frac{mv}{qB}.$$

[1 + 2 = 3 marks]

(a) (i) Clockwise

(ii) These two forces are equal, so

$$\frac{mv^2}{r} = qvB.$$

Cancelling, v gives

$$r = \frac{mv}{qB}.$$

Question 9 **10 marks**

(a)

(i) What work does the magnetic force do on a charged particle that moves through a uniform magnetic field. Explain your answer

(ii) A positive charged particle carries a charge of 0.2 μC and moves with a kinetic energy of 0.09 J. It travels through a uniform magnetic field with B = 0.1 T. What is the mass of the particle if it moves in the magnetic field in a circular manner with a radius of 3.0 m?

(b) State Faraday's law of induction. Describe two experimental arrangements which demonstrate this law.

[(1 + 4) + 5 = 10 marks]

(a) (i) None, because the force and the displacement are perpendicular.
1 marks

(ii) If the positively charged particle moves in the magnetic field in a circular fashion, the radius of the circular path is given in Eq. [20.15]. The kinetic energy of the particle is given as 0.09 J, so we know that this equals ½ m v^2. We can rearrange this formula for the mass and then substitute this into Eq. [20.15] to get v =3.0 × 10^6 m/s. Therefore, the particle has a mass given by m = 2.0 × 10^{-13} kg.

$Bqv = mv^2/R$, so, and $v = 2Ek/BqR$

(b) A change of Magnetic Flux through a loop gives rise to a current/voltage/electric field. 1

Any of the following

1. Pushing a magnet into a loop induces a current in the loop
2. Pulling a magnet away form a loop induced a current in the loop
3. Turn a coil in a magnetic filed and generate a current in the loop
4. Change the magnetic field by changing the current in an adjacent solenoid will generate a current in the loop.
5. Rapid change of magnetic field will generate an electric field, eg transcranial stimulation.

Question 10 **16 marks**

Two identical small conducting spheres are placed with their centres 0.300 m apart. One is given a charge of 12.0 nC and the other a charge of -18.0 nC.

(a) What is value of the magnitude of the electrostatic force exerted by each sphere on the other?

Since the spheres are 'small', we can assume that the force between them is given by the Coulomb force law:

$$\vec{F} = k_e \frac{q_1 q_2}{r^2} \hat{r}$$

In this case

$$q_1 = 12.0\ nC = 12.0 \times 10^{-9} C$$
$$q_2 = -18.0\ nC = -18.0 \times 10^{-9} C$$

and $r = 0.300\ m$.

The constant k_e is provided in the equation sheet: k_e = 8.988 × 10^9 N m²/C².

The force between the two spheres is *attractive*.

Substituting the values provided into the above expression gives:

$$F = 8.988 \times 10^9 \times \frac{12.0 \times 10^{-9} \times 18.0 \times 10^{-9}}{0.300^2}$$
$$= 8.988 \times \frac{12.0 \times 18.0}{0.0900} \times 10^{-9} = 21.57 \times 10^3 \times 10^{-9}$$
$$= 21.6 \times 10^{-6} N = 21.6 \mu N$$

(b) What is the magnitude and direction of the electric field at a point, P, midway between the two spheres? In your answer ensure you include a clear diagram indicating the direction of the field relative to the locations of the charges.

The electric field midway between the two spheres will be the Vector superposition of the electric fields produced by two point charges.

$$\vec{E} = k_e \frac{q}{r^2}\hat{r}$$

The contribution to the field at P from the positive charge will (according to the diagram) be directed to the right. Similarly, the contribution from the negative charge will be directed toward it and also to the right. Hence,

$$E = E_+ + E_-$$

$$E_+ = 8.988 \times 10^9 \times \frac{12.0 \times 10^{-9}}{0.150^2} = 8.988 \times \frac{12.0}{0.0225} = 4.79 \times 10^3 N/C$$

$$E_- = 8.988 \times 10^9 \times \frac{18.0 \times 10^{-9}}{0.150^2} = 8.988 \times \frac{18.0}{0.0225} = 7.19 \times 10^3 N/C$$

$$\therefore E = E_+ + E_- = 11.98 \times 10^3 N/C = 12.0 \times 10^3 N/C$$

to 3 significant figures.

The field is directed to the *right* according to the figure drawn above.

(c) What is the value of the electric potential at P assuming that the potential far from P is equal to zero?

The electric potential will be the scalar sum of the potential at this point due to each of the charges separately:

$$V = V_+ + V_-$$

Assuming the electric potential is defined to be zero at infinity:

$$V = k_e \frac{q}{r}$$

(d) How much work must to be done to move a third small charge of - 12.0 nC to this point P from a location far from P?

$$V_+ = 8.988 \times 10^9 \times \frac{12.0 \times 10^{-9}}{0.15} = \frac{8.988 \times 12.0}{0.15} = 719.0\ V$$

$$V_- = -8.988 \times 10^9 \times \frac{18.0 \times 10^{-9}}{0.15} = \frac{8.988 \times 18.0}{0.15} = -1079\ V$$

$$\therefore V(P) = V_+ + V_- = -360\ V$$

(e) The charge at P is removed and the two spheres, which remain in place, are then connected by a conducting wire. After the system reaches electrostatic equilibrium, the wire is removed.

(i) What is the magnitude of the electrostatic force each sphere exerts on the other?

(i)	If the spheres are connected by a conducting wire and allowed to come to electrostatic equilibrium, since they are identical, both spheres will have an identical charge: $$q = \frac{(q_1+q_2)}{2} = \frac{(12-18)}{2} = -3.00C$$ There will be a force of $$F = k_e \frac{q^2}{r^2} = 8.988 \times 10^9 \times \frac{(3.0 \times 10^{-9})^2}{0.300^2} 8.988 \times \frac{9.00}{0.09} \times 10^{-9} N$$ $$\therefore F = 8.99 \times 10^{-7} N$$
(ii)	It will be repulsive

(ii) Is the force repulsive or attractive?

The amount of work done by the field in moving a point charge q through an electric potential difference will be:

$$W = -\Delta U = -q\Delta V$$

Now $q = -12.0\ nC = -12.0 \times 10^{-9} C$ and $\Delta V = -360\ V$.

This means that point P is at a lower electric potential than points far from it. Hence

$$W = -q\Delta V = -(-12.0 \times 10^{-9})(-360\ V) = -4.320 \times 10^{-6} J$$

This is the work done *by the field* and hence work of +4.32 μJ will need to be done by an external agent to move the charge to point P.

[3 + 4 + 3 + 3 + 3 = 16 marks]

Question 11 **4 marks**

(a) A negatively charged particle is injected into a uniform magnetic field. The direction in which the particle is travelling is perpendicular to the direction of the field.

(i) Describe the motion of the particle.

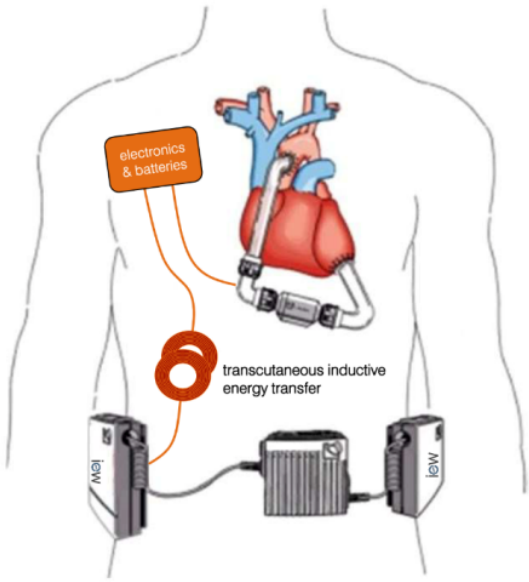
(ii) If the particle is injected into a uniform magnetic field in an arbitrary direction, discuss how your answer to part (i) changes.

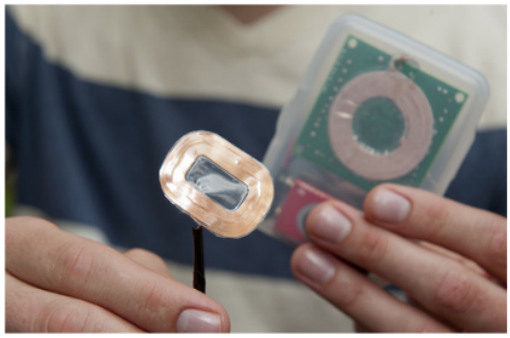
(i)	This particle will undergo circular motion.
(ii)	In this case, the component of the velocity vector parallel to the magnetic field will not be affected by the magnetic field, but the component perpendicular to the field will vary as in part (i). Hence, the overall motion will be a superposition of uniform motion parallel to the magnetic field and circular motion – helical motion.

[1 + 3 = 4 marks]

Question 12 **16 marks**

Medical devices implanted inside the body can be powered using transcutaneous energy transfer (TET) a type of wireless charging using a pair of closely spaced coils.





Consider a system where the coils are identical, each consisting of 10 turns and having a radius of 1.50 cm. The current in the external coil changes linearly from its maximum value of 10.0 A to zero in 6.25 μs.

(a) What is the maximum value of the magnetic field in the centre of the primary (external) coil?

The magnetic field at the centre of a single turn of a wire is given by:

$$B_{centre} = \frac{\mu_0 I}{2R}$$

Here

$$I_{max} = 10.0\ A$$
$$R = 1.50\ cm = 0.015m$$

and

$$\mu_0 = 4\pi \times 10^{-7}\ Tm/A$$

So for a single turn of wire:

$$B_{centre} = \frac{4\pi \times 10^{-7} \times 10.0}{2 \times 0.015} = 4.189 \times 10^3 \times 10^{-7} = 4.19 \times 10^{-4} T$$

In this case, there are 10 turns, so this increases the field by a factor of 10,

$$B_{max} = 4.19 \times 10^{-3} T = 4.19\ mT$$

(b) Assume that the magnetic field has approximately the same magnitude and direction at all points inside the primary coil as it has at the centre of the coil. What is the maximum value of the magnetic flux through the primary coil?

The maximum value of the magnetic flux will be

$$\Phi_{max} = B_{max}A = B_{max}\pi R^2 = 4.19 \times 10^{-3}T \times \pi \times 0.015^2 m^2 = 0.00296 \times 10^{-3}Wb$$

$$\Phi_{max} \approx 2.96 \times 10^{-6}Wb$$

The two coils are parallel and located close to each other, so that the flux through the implanted (secondary) coil can be taken to be identical to that through the external coil.

(c) What is the value of the induced emf E in the implanted coil?

From Faraday's Law of Induction the induced emf is given by:

$$emf = -\frac{d\Phi_B}{dt}$$

We know that the flux decreases linearly to zero in 6.25 μs, so

$$emf = \frac{|\Delta\Phi_B|}{\Delta t}\frac{2.96 \times 10^{-6}}{6.25 \times 10^{-6}} = 0.474\ V$$

(d) If the current in the external coil is clockwise, what will be the direction of the induced current in the implanted coil? Explain.

The flux will be decreasing, so the induced current will be such as to oppose this decrease in flux. The induced current will also be *clockwise* since this produces a magnetic field in the same direction as that produced by the primary and will tend to reinforce it.

(e) If the internal coil has a resistance of 2.8 W

(i) What is the value of the current in the coil?

(ii) What is the value of the power dissipated in the coil?

(i)	$I = \frac{\mathcal{E}}{R} = \frac{0.474}{2.8} = 0.17\ A$
(ii)	$P = I\mathcal{E} = 0.169 \times 0.474 = 0.080A = 80\ mA$

[4 + 3 + 3 + 2 + 4 = 16 marks]